Nexans

ULTREX® VN

PVC/Nylon Type TC 600 Volt Power and Control Cable Temperature rating of 90°C dry / 75°C wet rated - Lead Free

Applications

Nexans 600V ULTREX® VN Tray Cables are listed as type TC under UL 1277 Electrical Power and Control Cables. These cables may be installed in wet or dry locations; in cable trays, raceways and open air; and are suitable for exposure to weather, direct burial and for Class I, Div. 2 (also Zone 2) and Class II, Div. 2 hazardous locations per NEC.

Construction

Conductor: bare, annealed copper conforming to ASTM B-3 and B-8, from #18 AWG to 500 kcmil. Compressed copper for #14 AWG through 500 kcmil.

Insulation: flame-retardant PVC/Nylon type THHN/THWN per UL 83 for sizes #14 AWG to 500 kcmil and type TFN per UL 66 for #18 AWG and #16 AWG.

Assembly:
Non-shielded: cables with 3 or more conductors are cabled in concentric layers with interstices filled with suitable fillers, as required. Two-conductor cables are supplied in a flat/parallel configuration. Bare grounds, when provided, are sized as required by UL 1277 (refer to the applicable product table(s) for the standard sizes provided). Where necessary, a binder tape of synthetic material assembles the core in a tight circular configuration.
Shielded: cabled in concentric layers with interstices filled with suitable fillers as required. A helically wrapped aluminum tape, with synthetic backing, gives 100% shielding. A tinned copper drain wire is placed in contact with the aluminum side of the tape, to lower the resistance and to assist in the termination of the shield.

Jacket: UL listed sunlight and moisture resistant, sequentially length marked, black, flame retardant polyvinyl chloride (PVC) material meeting the requirements of UL 1277. A Nylon ripcord is included for ease of jacket removal.

Conductor Identification

(See page 26 of this catalog)

#18 AWG to #10 AWG: color coded per Method #1-E2 per ICEA S-73-532

#8 AWG to 500 kcmil: black with number coding per Method 4 of ICEA S-73-532

Composite
Power and Control: black with number coding per Method 4 of ICEA S-73-532

Specifications

- Meets UL 1277: Power and Control Tray Cables with Optional Fiber Members.
- Meets UL 83: Thermoplastic-Insulated Wires and Cables for #14 AWG and larger.
- Meets UL 66: Fixture Wire for #18 AWG and #16 AWG.
- Meets ICEA S-95-658, NEMA Publication No. WC-70: Non-Shielded Power Cables rated 2000 volts or Less for the Distribution of Electrical Energy.

Product Features

- UL approved cables Type TC, 600V.
- UL approved insulated conductors.
- Cables pass UL 1685 and IEEE 383 vertical tray fire tests at 70,000 BTU/hr.
- Cables are UL listed to IEEE 1202 and FT4 70,000 BTU/hr flame test.
- For use in power, lighting, control and signal circuits.
- Can be used within industrial establishments where serviced by qualified personnel and not subject to physical damage.
- Can operate at continuous temperatures of 90°C dry and 75°C wet and cold bend of –25°C per UL 1277 Section 15.
- Can be used In Class I Division 2 and Class II Division 2 Hazardous Locations and Intrinsically Safe applications as permitted by NEC Art. 392, 501, 502, 503, and 505.*
- As indicated in UL 1277: The overall jackets of these cables are a "gas/vapour tight continuous sheath" as discussed in NEC Sections 501.15(D) & (E).*
- For use in cable trays, raceways, conduits, or for aerial applications where installed with a messenger.
- UL listed for Exposed Run (ER) use for external runs per NEC Article 336.10(7) for sizes 2/0 and larger. (4 AWG–1/0 AWG upon request.)
- For Direct Burial applications.
- As permitted in NEC Art. 725 for Class 1 circuits.
- As permitted for non-power-limited fire alarm circuits as defined in NEC Article 760
- Lead free and RoHS compliant
- Sunlight and moisture resistant

Options

The following constructions can be provided on special orders:
- Dow Construction
- DuPont Construction
- Composite Cable Construction
- Insulated or bare ground wires
- Different conductor identification methods

* Use in hazardous locations:
Please note that no investigation of these cables has been performed regarding the transmission of gases or vapours through the core. When these cables are used in hazardous locations they should be sealed properly as required by the NEC.

Unshielded Tray Cable
UL Type TC, 600V, 90°C dry / 75°C wet rated - Lead Free

4 Conductors with Bare Ground Type THHN/THWN

Part Number	Conductor Size	Ground Size	Insulation Thickness PVC/Nylon		Jacket Thickness		Nominal Diameter over Jacket		Approximate Net Cable Weight		Ampacity [1,3,4]	
	AWG/ kcmil	AWG	mils	mm	mils	mm	inches	mm	lb/kft	kg/km	amps	amps[2]
697607	8	10	30/5	.76/.13	60	1.52	.661	16.79	353	525	44	55
697599	6	8	30/5	.76/.13	60	1.52	.746	18.95	516	768	60	75
697581	4	8	40/6	1.02/.15	80	2.03	.957	24.31	804	1196	76	95
697573	2	6	40/6	1.02/.15	80	2.03	1.102	27.99	1108	1649	104	130
———	1	6	50/7	1.27/.18	80	2.03	1.247	31.67	1519	2260	120	150
669135	1/0	6	50/7	1.27/.18	80	2.03	1.344	34.14	1802	2682	136	170
669143*†	2/0	6	50/7	1.27/.18	80	2.03	1.452	36.88	2288	3405	156	195
———	3/0	4	50/7	1.27/.18	80	2.03	1.575	40.01	2756	4101	180	225
669279*†	4/0	4	50/7	1.27/.18	110	2.79	1.768	44.91	3437	5115	208	260
———	250	4	60/8	1.52/.20	110	2.79	1.937	49.20	4046	6021	232	290
669283*†	350	3	60/8	1.52/.20	110	2.79	2.180	55.37	5480	8155	280	350
669291*†	500	2	60/8	1.52/.20	110	2.79	2.494	63.35	7574	11270	344	430

* DOW construction cables. (See page 18)

† ER approved for exposed runs per NEC 336.10(7)

Bend Radius: 5 x overall diameter installed (up to 2.0") / 6 x overall diameter installed (over 2.0")
8 x overall diameter during installation pull-in.

Notes:

Dimensions and weights shown are nominal values. They are subject to standard industry tolerances.
Cables with different conductor counts and bare or insulated grounds are also available.

(1) Ampacities are in accordance with NEC Table 310.16 for conductors in a raceway or direct buried at 30°0C ambient temperature and 90°C rated conductors. Ampacities for cables having more than three conductors have been derated per NEC Table 310.15(B)(2)(a).

(2) Where the 4th conductor is the neutral of a balanced 3 phase system.

(3) For load diversity of 50%, refer to NEC Table B.310.11.

(4) For correction factors to different ambient temperatures and ampacities for different conductor temperature ratings see NEC Table 310.16.